

Report: Delegation & Least Privilege

1. Introduction

Delegation of control in Active Directory allows administrative rights to be assigned in a restricted and controlled manner to specific groups without giving full domain-wide privileges. Least privilege ensures that accounts or users are only granted permissions that are strictly necessary for their tasks. In this section, delegation was used to allow a Helpdesk group to reset passwords for users inside the Departments OU. This prevents giving unnecessary administrative access to helpdesk personnel. The delegation was followed by testing to confirm it worked as expected. After this, an audit was performed on the "Domain Admins" group to ensure only necessary users were retained, and unnecessary users were removed.

2. Objective

The objective of this task group was to implement secure, role-based access through delegation and apply the principle of least privilege across the environment. Specifically, a group named Helpdesk was to be granted the ability to reset passwords for users under the Departments OU. After delegation, a test was to be performed to validate that this permission worked as expected using a Helpdesk-assigned user account. Following successful validation, the final task involved checking the Domain Admins group and ensuring that only intended users (like Administrator) were retained, and any unnecessary user accounts were removed. This was done to reduce administrative exposure and enforce minimum access.

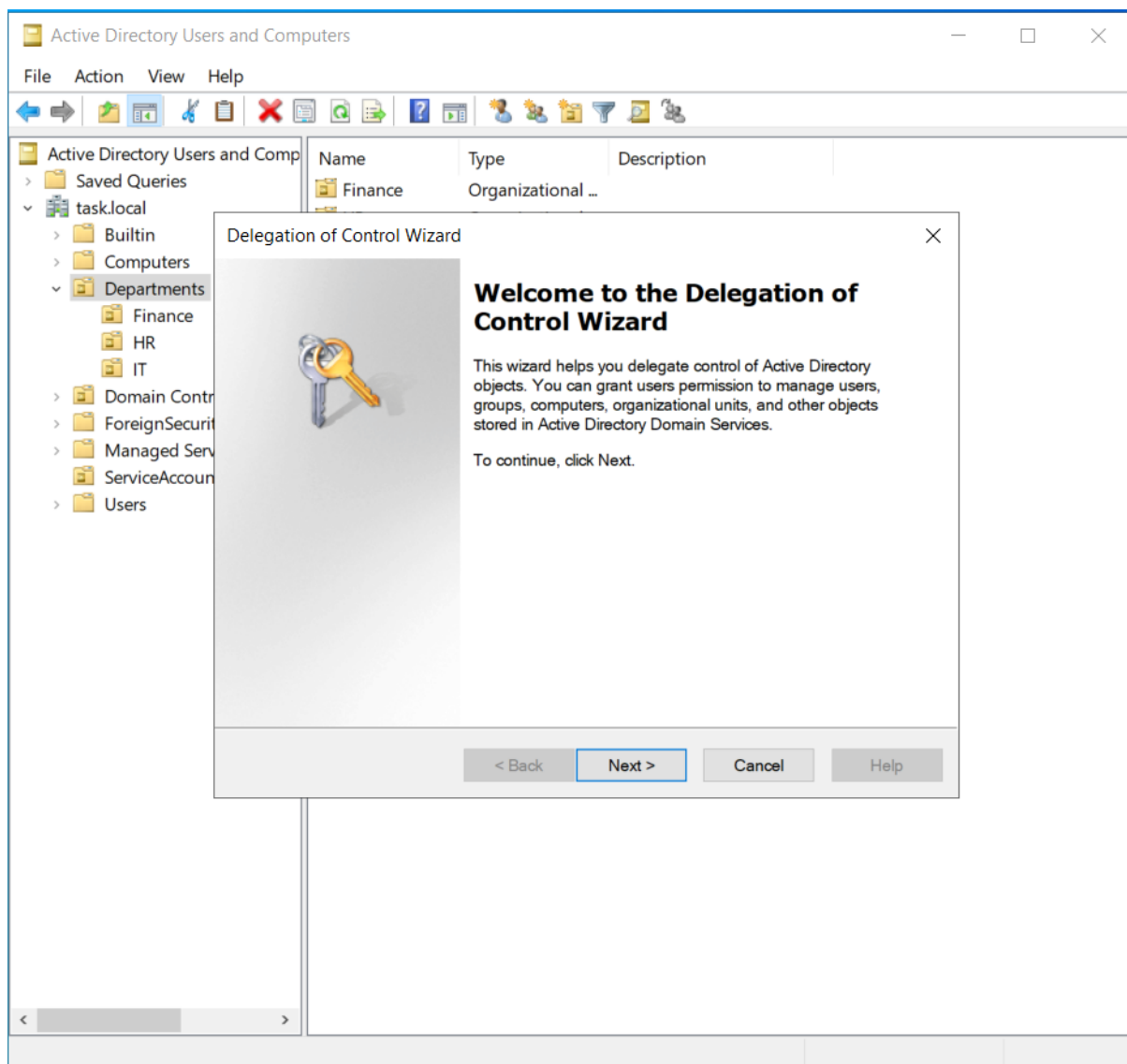
3. Tools used

- Active Directory Users and Computers (ADUC)
- RSAT (Remote Server Administration Tools)
- Windows PowerShell

4. Methodology

4.1 Delegate "Reset Password" Rights to the Helpdesk Group

A security group named Helpdesk was first created under the Departments OU using ADUC. After group creation, the Delegation of Control Wizard was launched on the Departments OU. During delegation setup, the Helpdesk group was selected as the target group. In the wizard options, the task “Reset user passwords and force password change at next logon” was selected. Once completed, the delegation was in effect and ready to be tested.



(Run delegation wizard on departments)

Delegation of Control Wizard



Tasks to Delegate

You can select common tasks or customize your own.



☒ Delegate the following common tasks:

☐ Create, delete, and manage user accounts	^
☒ Reset user passwords and force password change at next logon	
☐ Read all user information	
☐ Create, delete and manage groups	
☐ Modify the membership of a group	
☐ Manage Group Policy links	
☐ Generate Resultant Set of Policy (Planning)	v
<	>

☐ Create a custom task to delegate

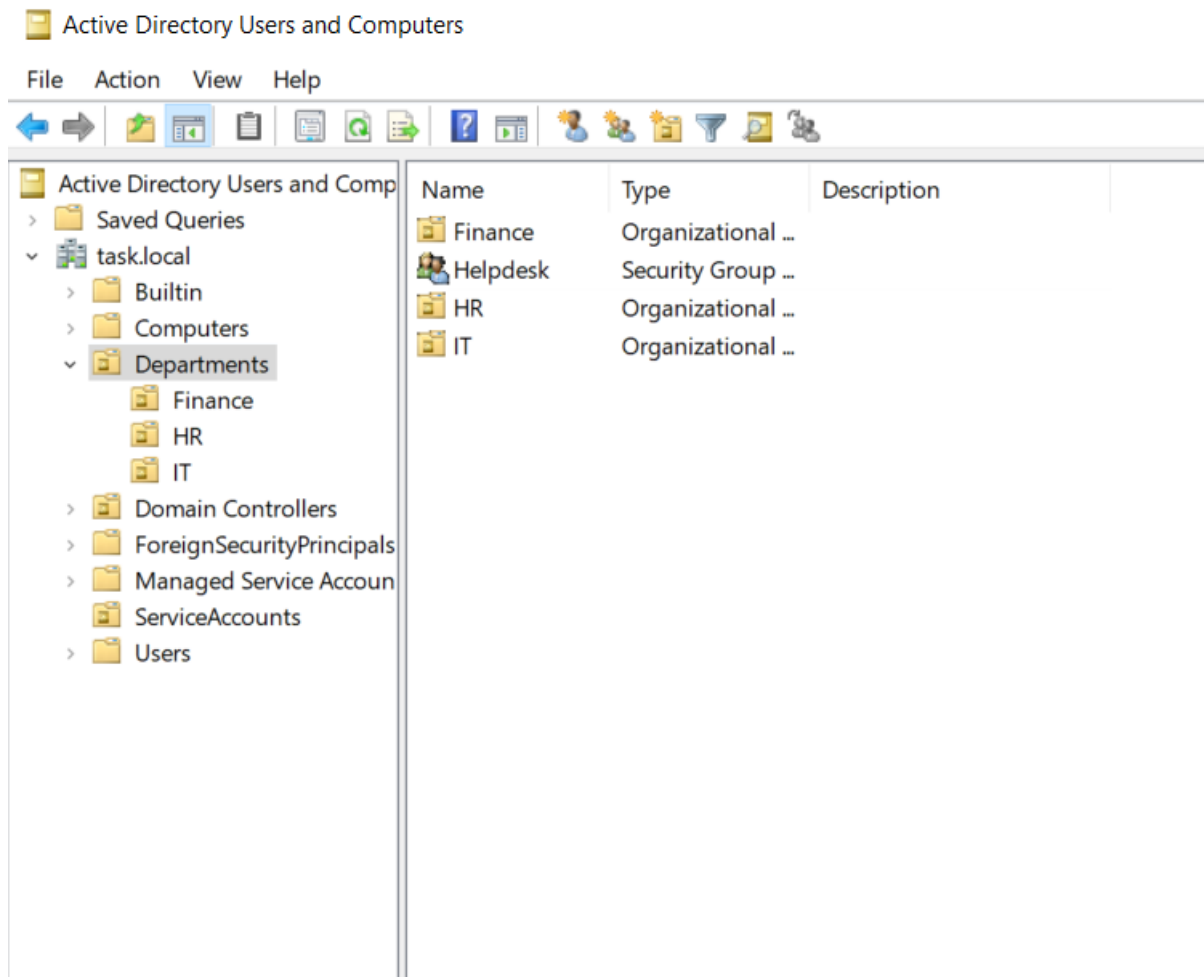
< Back

Next >

Cancel

Help

(Final confirmation screen of the delegation wizard)

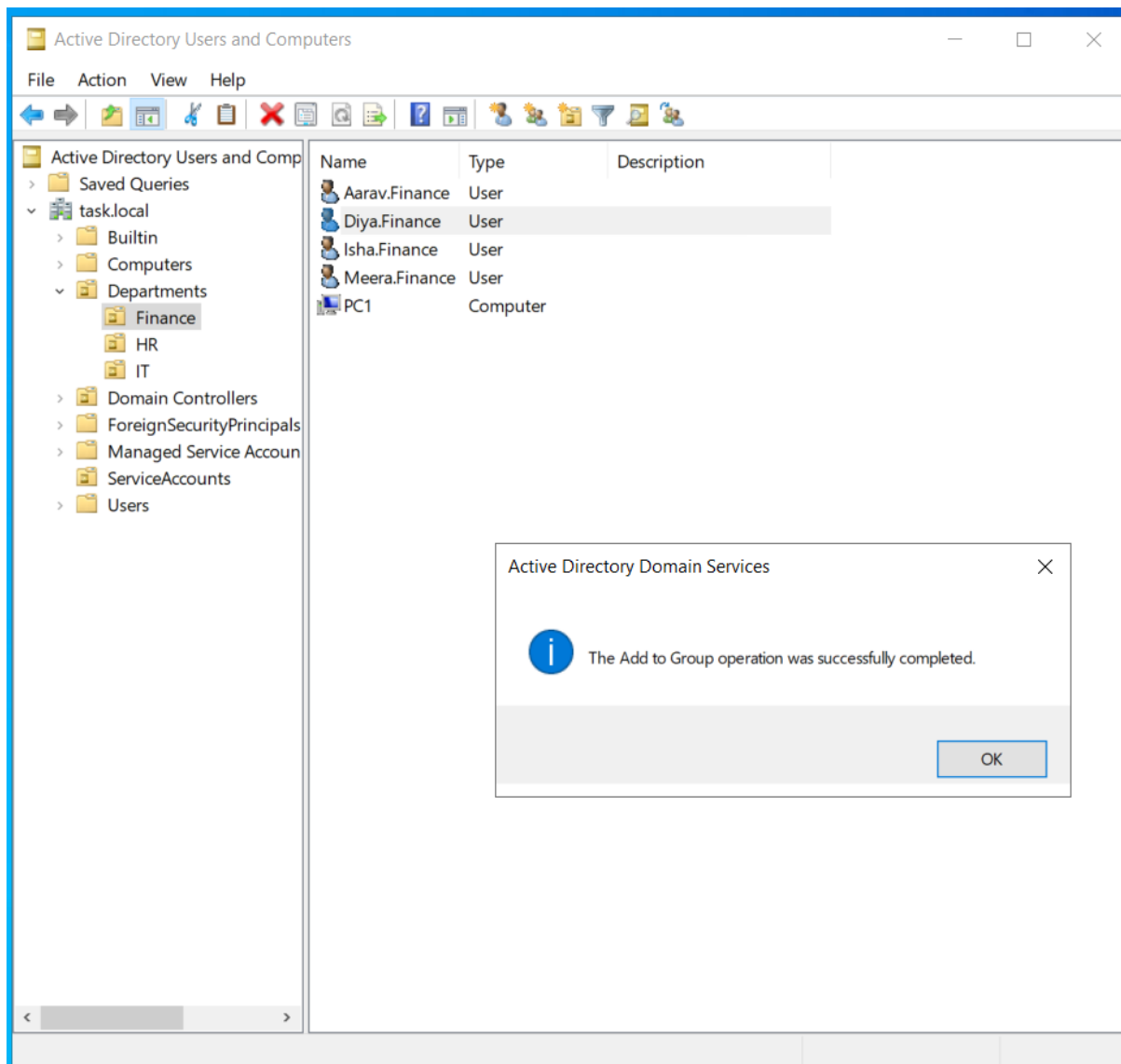


(Now we have helpdesk with right to reset user passwords and force password change at next logon)

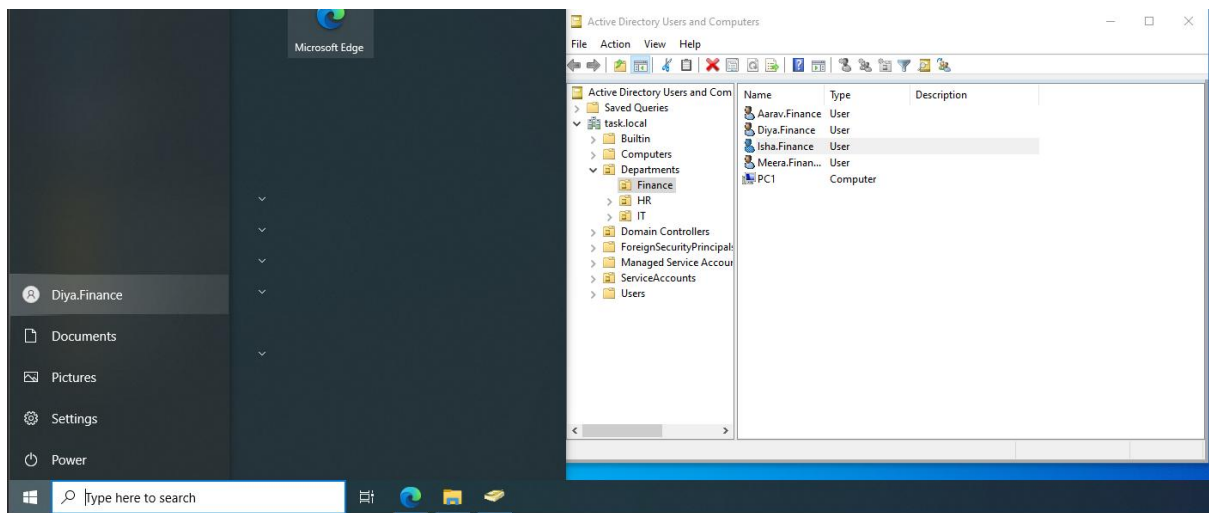
4.2 Test Delegated Permissions by Resetting a Test Account Password

To validate the delegation, a user named Diya.Finance was added to the Helpdesk group. RSAT was installed on a Windows 10 domain-joined client using PowerShell. Once RSAT was confirmed as installed, dsa.msc was executed and ADUC was opened successfully. Logged in as Diya.Finance, navigation was done to the Departments > Finance OU. Inside that OU, the user Isha.Finance was selected and the Reset Password option was used. A new password was entered and the password reset completed

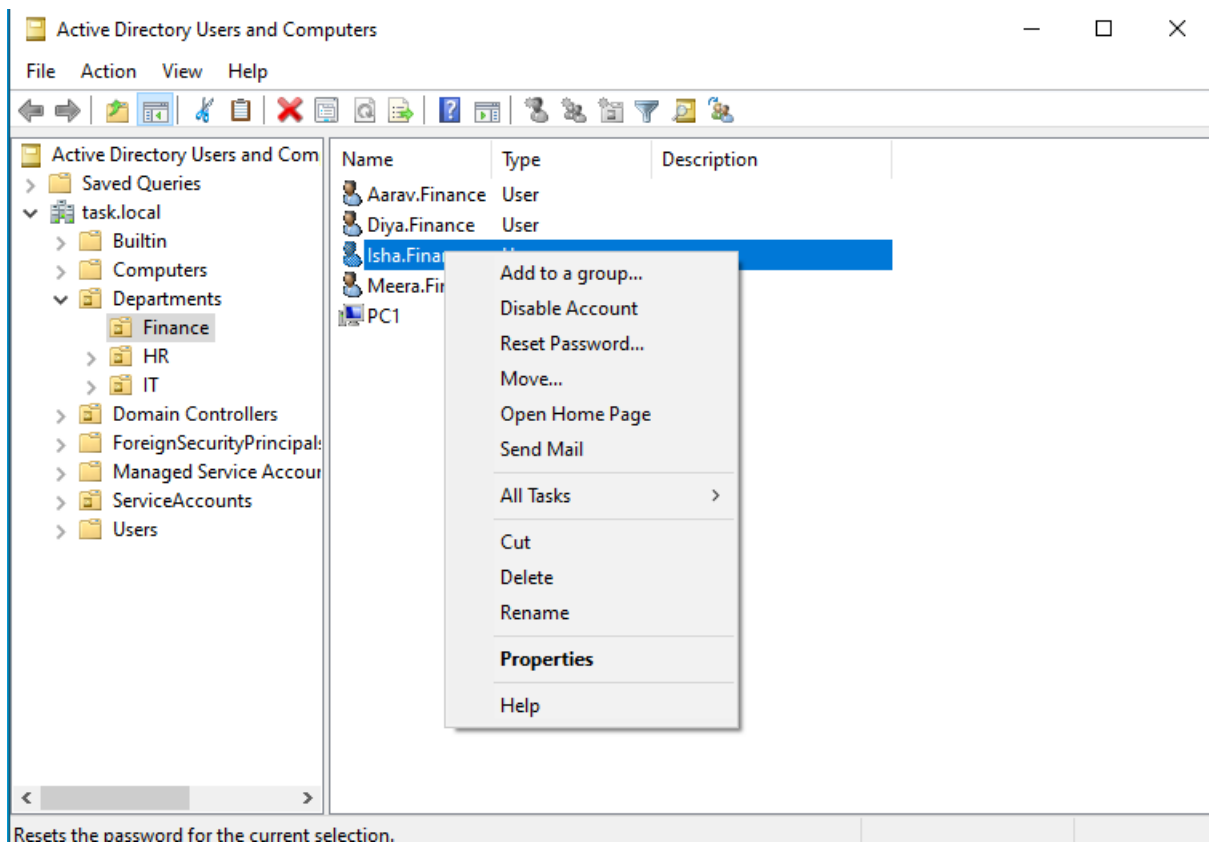
without any error. This confirmed that the delegated permissions worked correctly and Helpdesk members were able to reset user passwords as intended.



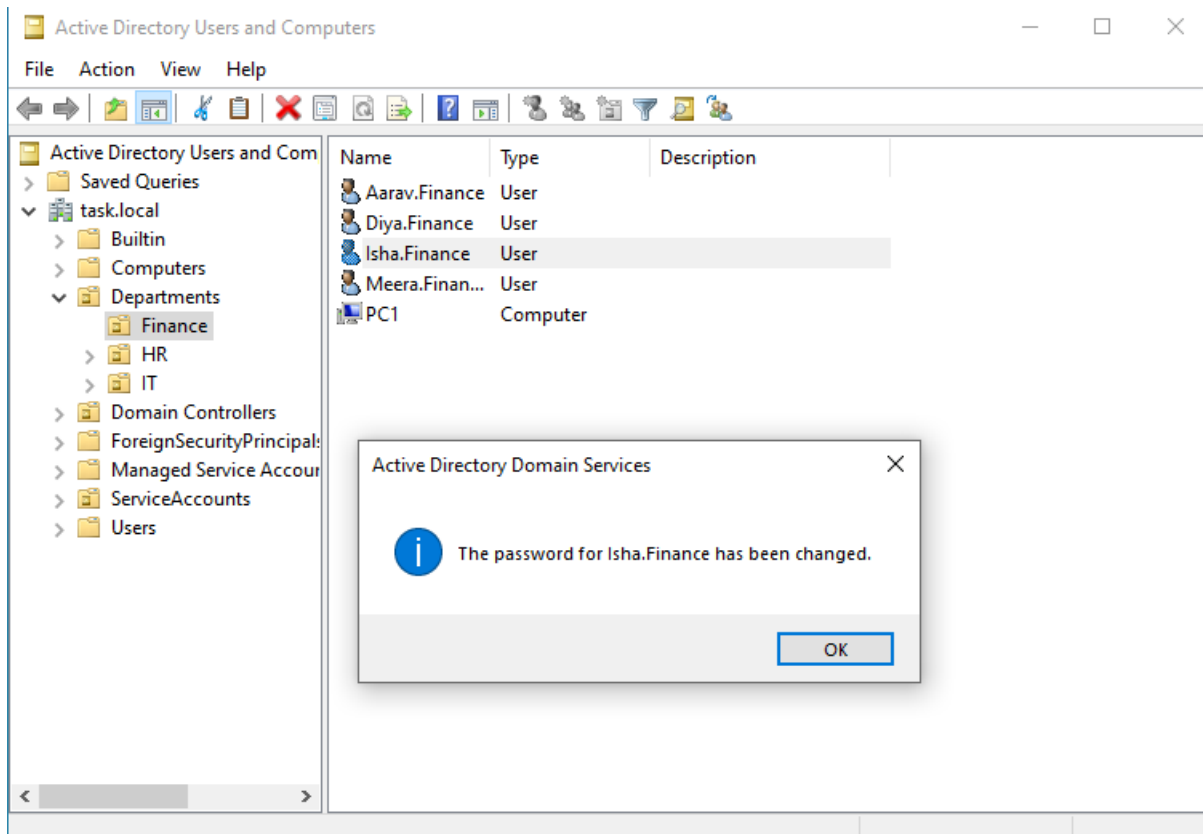
(Assigned Diya.Finance to helpdesk group)



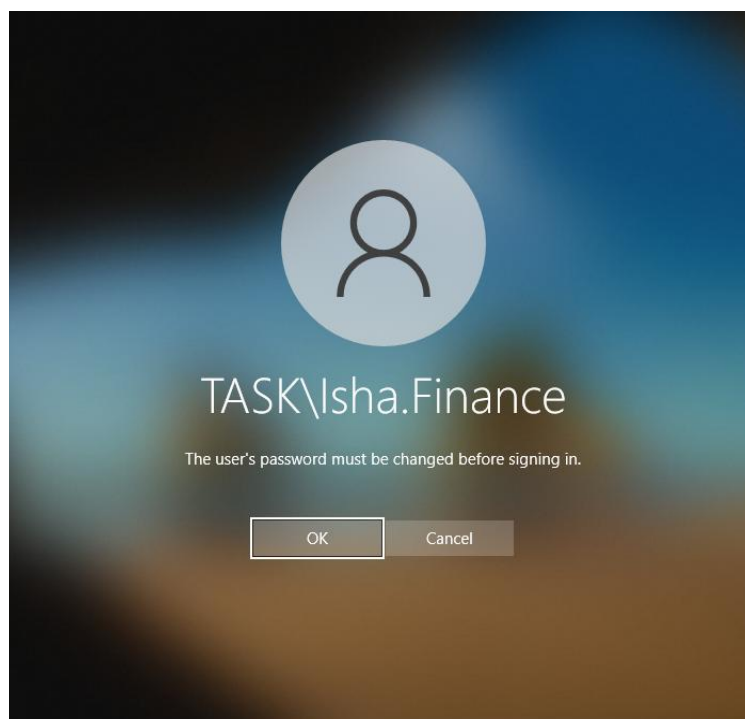
(Login as Diya.Finance)



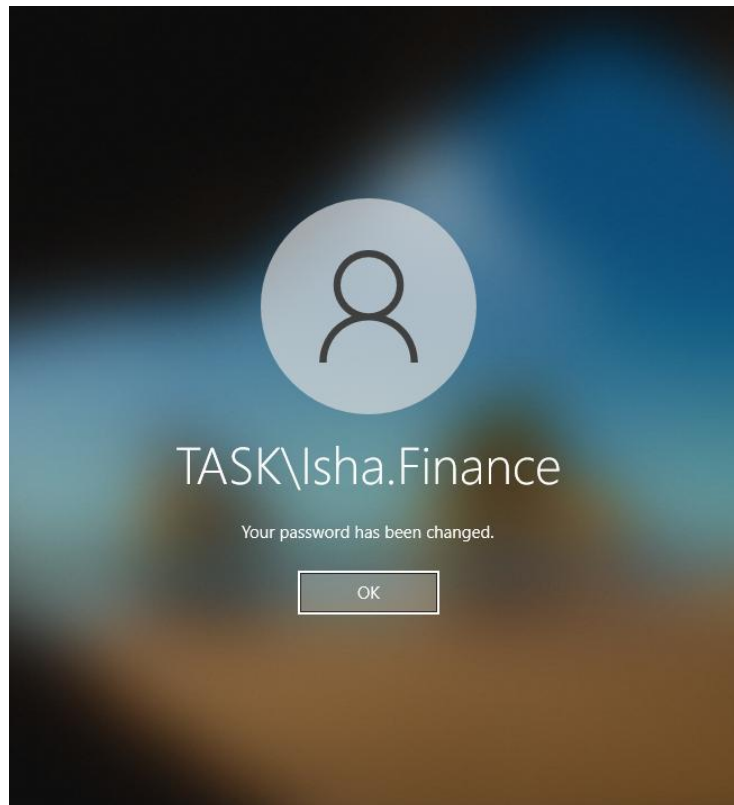
(Reset password of Isha.Finance)



(Command to reset password from helpdesk user diya has been executed)



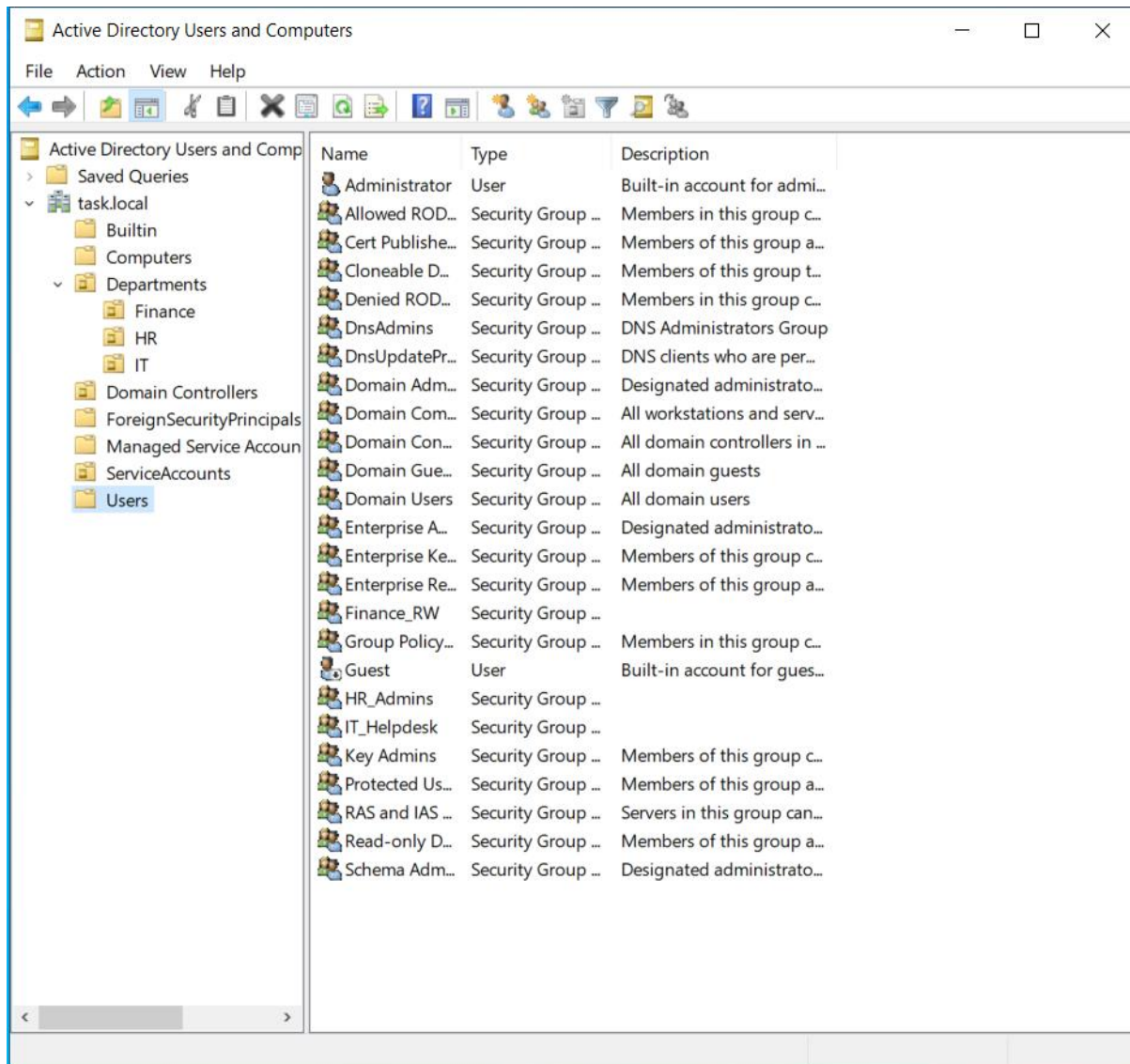
(We have to change the password before login)



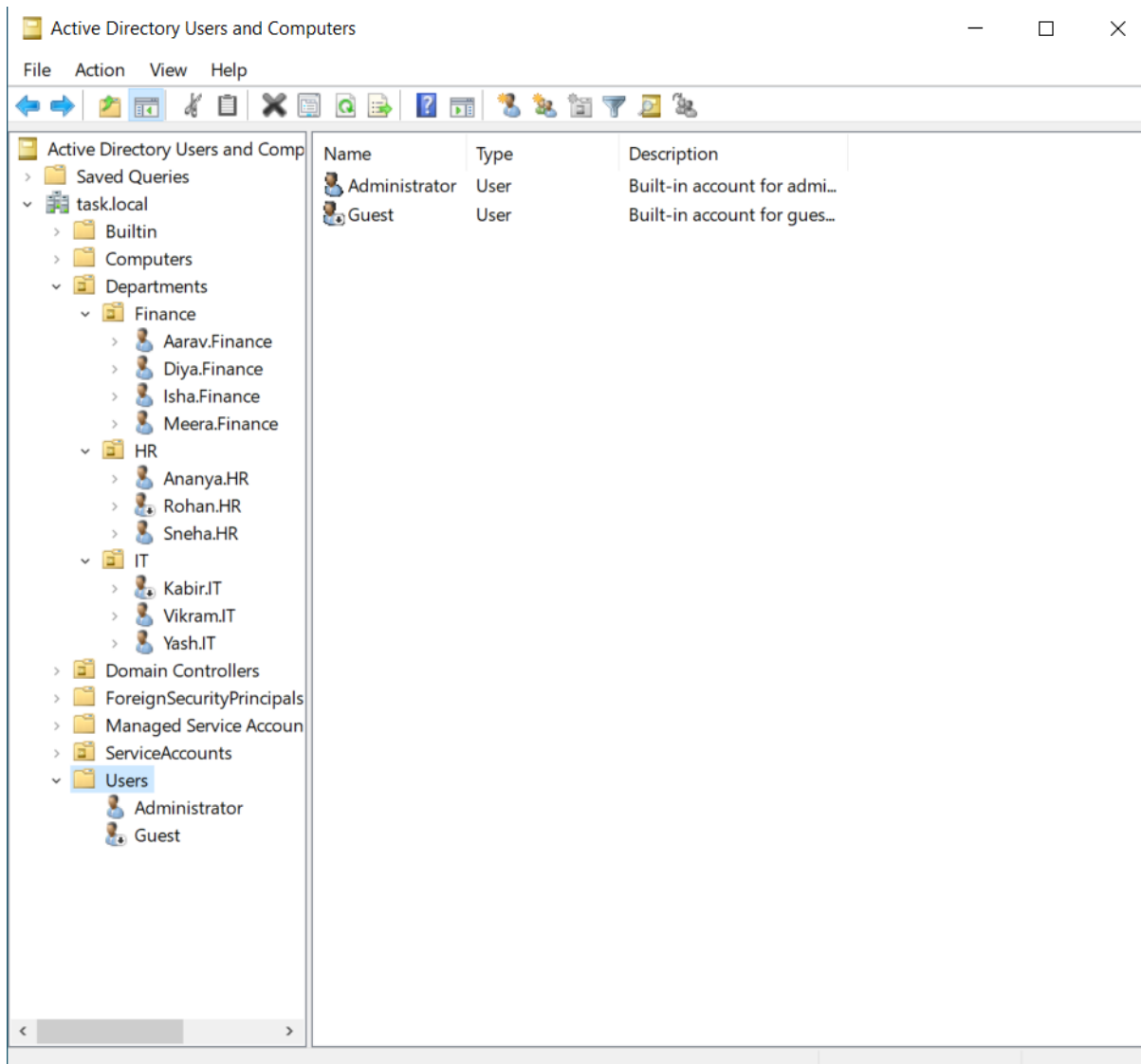
(Changed the password successfully)

4.3 Remove Unnecessary Users from "Domain Admins" Group

Using ADUC on the Windows Server (Domain Controller), navigation was done to the Users container. Inside this container, the Domain Admins group was located and opened. Under the "Members" tab, the listed users were reviewed. The group contained only the Administrator account and the default Guest account in the container. No unnecessary users were found as members of the Domain Admins group. Since no other helpdesk or departmental users were part of this group, no further action was required and the task was considered complete.



(In ADUC click on users and check which user you want to remove)



(There are no unnecessary users remaining in the Domain Admins group if only the Administrator account is a member.)

5. Recommendations

- Perform delegation only on specific Organizational Units (OUs) where necessary.
- Use the Delegation of Control Wizard to assign only the required permissions (e.g., "Reset password").
- Avoid assigning full domain-wide privileges to Helpdesk or departmental users.
- Regularly review and audit memberships of privileged groups such as Domain Admins.

- Ensure that no unnecessary or temporary accounts are present in high-privilege groups.
- Use clearly named groups (e.g., Helpdesk) for assigned roles to maintain clarity and manageability.
- Install RSAT only on trusted and managed machines for delegated administrative use.
- Test delegated permissions using non-admin accounts before putting them into active use.

6. Conclusion

Delegated permissions were successfully configured to allow the Helpdesk group to reset passwords for user accounts inside the Departments OU. The group was tested by assigning the user Diya.Finance to Helpdesk and using that account to reset the password for Isha.Finance, which worked as intended. No additional permissions were given beyond what was delegated. An audit of the Domain Admins group confirmed that only the default administrative account was present. No extra or non-administrative users were found in the group. This completes all tasks under Delegation & Least Privilege.